MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **PERIODIC TEST – 02 (SURFACE AREAS & VOLUMES)** | **Time:** 60 Minutes

Class: Class X (CBSE) | **Chapter 13: Surface Areas and Volumes** | **MM: 30**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Section A has 4 MCQs and 1 Assertion-Reason question (1 mark each).
2. Section B has 2 VSAQ questions (2 marks each). Section C has 2 SA questions (3 marks each).
3. Section D has 1 Long Answer HOTS question (5 marks). Section E has 1 Applied Case Study (4 marks).
4. Take $\pi = \frac{22}{7}$ unless specified otherwise. Show clear steps in calculations.

SECTION A: OBJECTIVE & MCQs (1 Mark Each) [5 Marks]

1. The maximum volume of a cone that can be carved out of a solid hemisphere of radius r is: **[1]**

(a) (a) $3\pi r^2$ (b) $\frac{1}{3}\pi r^3$ (c) $\frac{2}{3}\pi r^3$ (d) πr^3

2. The radii of two cylinders are in the ratio 3 : 5 and their heights are in the ratio 2 : 3. The ratio of their curved surface areas is: **[1]**

(a) (a) 2 : 5 (b) 5 : 2 (c) 3 : 5 (d) 6 : 25

3. If the radius of the base of a right circular cylinder is halved, keeping height the same, the ratio of the volume of reduced cylinder to the original cylinder is: **[1]**

(a) (a) 1 : 2 (b) 1 : 4 (c) 1 : 8 (d) 4 : 1

4. The material of a cone is converted into the shape of a cylinder of equal radius. If the height of the cylinder is 5 cm, the height of the cone was: **[1]**

(a) (a) 10 cm (b) 15 cm (c) 18 cm (d) 24 cm

5. **Assertion (A):** A conical vessel of base radius 5 cm and height 24 cm is full of water. If emptied into a cylindrical vessel of internal radius 10 cm, the water level rises by 2 cm.
Reason (R): The volume of water in the conical vessel is $\frac{1}{3}\pi r_1^2 h_1$, and the volume of water displaced in the cylinder is $\pi r_2^2 h_2$. **[1]**

(a) Both (A) and (R) are true, and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(c) (A) is true, but (R) is false.
(d) (A) is false, but (R) is true.

SECTION B: VERY SHORT ANSWER QUESTIONS (2 Marks Each) [4 Marks]

6. If a right circular cone and a sphere have equal radii and equal volumes, what is the ratio of the diameter of the sphere to the height of the cone? **[2]**

7. A cone, a hemisphere, and a cylinder stand on equal circular bases of radius r and have the same height r. What is the ratio of their volumes? **[2]**

SECTION C: SHORT ANSWER QUESTIONS (3 Marks Each) [6 Marks]

8. A solid is in the shape of a right circular cone surmounted on a hemisphere, the radius of each of them being 7 cm and the height of the cone is equal to its diameter. Find the volume of the solid. **[3]**

9. Two solid cones A and B are placed inside a cylindrical tube of radius 3 cm and height 21 cm. The ratio of their capacities is 2 : 1. Find the heights and capacities of the cones, and the volume of the remaining portion of the cylinder. **[3]**

SECTION D: LONG ANSWER / HOTS QUESTION (5 Marks) [5 Marks]

10. A spherical glass vessel has a cylindrical neck 8 cm long and 2 cm in diameter; the diameter of the spherical portion is 8.5 cm. By measuring the amount of water it holds, a student finds its volume to be 345 cm^3. Check whether the student is correct, taking the inside measurements and $\pi = 3.14$. **[5]**

SECTION E: APPLIED CASE STUDY (4 Marks) [4 Marks]

11. Case Study: Architectural Marvel – The Atal Highway Tunnel

Atal Tunnel under Rohtang Pass is an engineering wonder. The cross-section of the tunnel is in the shape of a major circular segment resting on a horizontal roadway AB. The centre of the circular cross-section is O, the radius of the circular part is $R = 5\sqrt{2}$ metres, and $\angle AOB = 90°$. The total length of the tunnel is 9 km = 9000 metres.

(i) Find the horizontal width of the roadway inside the tunnel (length of chord AB).

(ii) Find the total cross-sectional area of the tunnel opening.

(iii) Find the perimeter of the cross-section of the tunnel.

(iv) Find the surface area of the inner circular wall of the tunnel over its entire 9 km length.

[4]

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— END OF QUESTION PAPER —

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